

CTRL-C2000 control card

Rev E0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | TMS320F28388D common controller for PV-P75/100/110 and DAB-D60

01 Power regulators

+24V -> LMR38020 5 V -> TPS62133 3.3 V; PG chain enables TPS62130 1.2 V and TPS7A2033 3.3 V analog (VDDIO first, then VDD and VDDA)

File: sheets/01_power.kicad_sch

02 Supervisors, reset, clock, boot

TPS3703 windows on 3.3 V and 3.3 V analog, UV supervisor on 1.2 V -> XRSn; 25 MHz CMOS clock into X1; boot straps GPIO72/GPIO84 (flash by default)

File: sheets/02_reset.kicad_sch

03 TMS320F28388D (ZWT 337-ball)

All 337 balls, pin table extracted from SPRSP14E (gen/data/f28388d_pins.csv); net per ball from the pin plan (gen/data/ctrl_c2000_pin_plan.csv)

File: sheets/03_mcu.kicad_sch

04 MCU supply decoupling

SPRSP14E 7.10.1.5: 0.1 uF per VDDIO/VDD3VFL/VDDOSC ball, 2.2 uF per VDDA ball, -22 uF total on VDD + one 56 R load, 22 uF per VREFHI (16-bit mode)

File: sheets/04_decoupling.kicad_sch

05 ADC reference, VMID, slow mux

REF5025E 2.5 V for VREFHIA-D; OPA2388 buffers VMID 1.25 V (diff-amp offset) and VBIAS 2.5 V (ID/NTC bias); TMUX1208 for 4 NTC + 4 ID; 24 V monitor; AGND star

File: sheets/05_adcref.kicad_sch

06 Fault receivers, enable, EN drivers

Cn_EN = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only while CHB_OK, MCU enable and RDY; EN drivers with readback; fail-safe FLT receivers

File: sheets/06_trip.kicad_sch

07 Cell port C1

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/07_cell1.kicad_sch

08 Cell port C2

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/08_cell2.kicad_sch

09 Cell port C3

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/09_cell3.kicad_sch

10 Cell port C4

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/10_cell4.kicad_sch

11 Port-board analog inputs

PORT contract 2x13: VA, VAX, VB, VBX, IA, IB, AUX1, AUX2 isolated-amplifier pairs, biased open-safe, through difference amp + RC + clamp; ID: +24V out through an eFuse

File: sheets/11_port.kicad_sch

12 SYS-IO-AUX connector

80-way SYS contract: +24 V logic, +24V_GD (channel B), CAN, SCI, SPI, I2C, DI/DO, fault latch, GATE_EN, CHB_OK, watchdog, reset, fans, Ethernet MDI + centre tap

File: sheets/12_sys.kicad_sch

13 Ethernet PHY DP83822I

MII to the CM EMAC (GPIO105-124, mux 14), own 25 MHz oscillator, 49.9 R MDI terminations to AVD; MDI pairs, LEDs and the centre-tap supply ETH_CT go to SYS

File: sheets/13_eth.kicad_sch

14 JTAG, UART, EEPROM, LEDs, SDFM

cTI-20 JTAG, CM debug UART, 24LC256 calibration EEPROM (0x57), status LEDs, ERRORSTS, 8-channel SDFM expansion header (ECO-02 / ECO-09)

File: sheets/14_debug.kicad_sch

Protection without firmware: FLT_LATCH_N -> INPUTXBAR1 -> TZ1; each cell's FAULT (INPUTXBAR2-5) and AN1 CMPSS window -> TRIP4/5/7/8; port OV (ADC PPB on VA/VB, CMPSS backup), open inputs (CMPSS low), IA/IB SC (CMPSS) and OC (PPB), P_ID (PPB) -> TRIP9 for every cell; E-stop -> INPUTXBAR6 -> TRIP6. CMPSS reference = VDAC = REF5025E; firmware nulls each channel at zero input at start-up; trip bands in the pin plan, asserted against the frozen earlier-platform snapshot gen/data/control_spec_platform1.json (D-044). EN pair = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only with CHB_OK, MCU enable and RDY; EN readback on GPIO100-103. +24V_GD feeds only the cell eFuses. Rails: 24 V -> 5 V (LMR38020) -> 3.3 V (TPS62133); its PG enables 1.2 V (TPS62130) and 3.3 V analog (TPS7A2033), so VDDIO leads VDD and VDDA (SPRSP14E table 7-1). Three TPS3703 hold XRSn (windows on 3.3 V and 3.3 V analog, UV on 1.2 V), every tolerance corner asserted. REF5025E runs from the analog rail, so VREFHI and VDAC never exceed VDDA. Pin plan with mux positions and trip bands: gen/data/ctrl_c2000_pin_plan.csv (checked against SPRSP14E table 6-1 on every build). Pin table: gen/data/f28388d_pins.csv (from the PDF). SYS side: inputs via 74LVC244A / 74LVC1G17 (loff); I2C pull-ups and EEPROM 0x57 here; ETH_CT = PHY AVD; PORT +24V via its own eFuse.

Not bench-validated. Values marked calculated are not simulated.
Sheet 1
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /
File: CTRL-C2000.kicad_sch
Title: CTRL-C2000 control card - Cover / Hierarchy

Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 1/15

24 V -> 5 V (LMR38020, 400 KHZ)

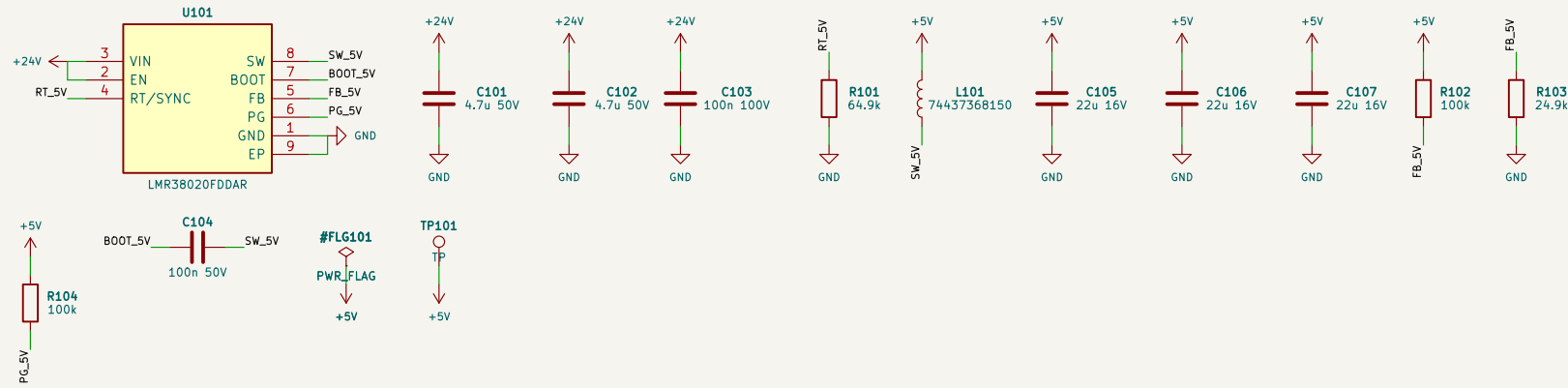
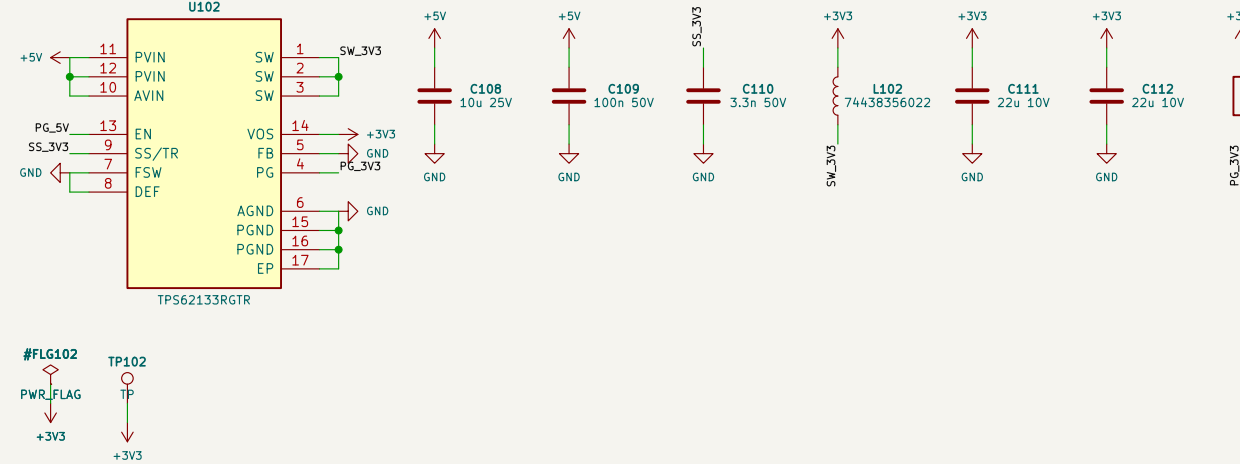


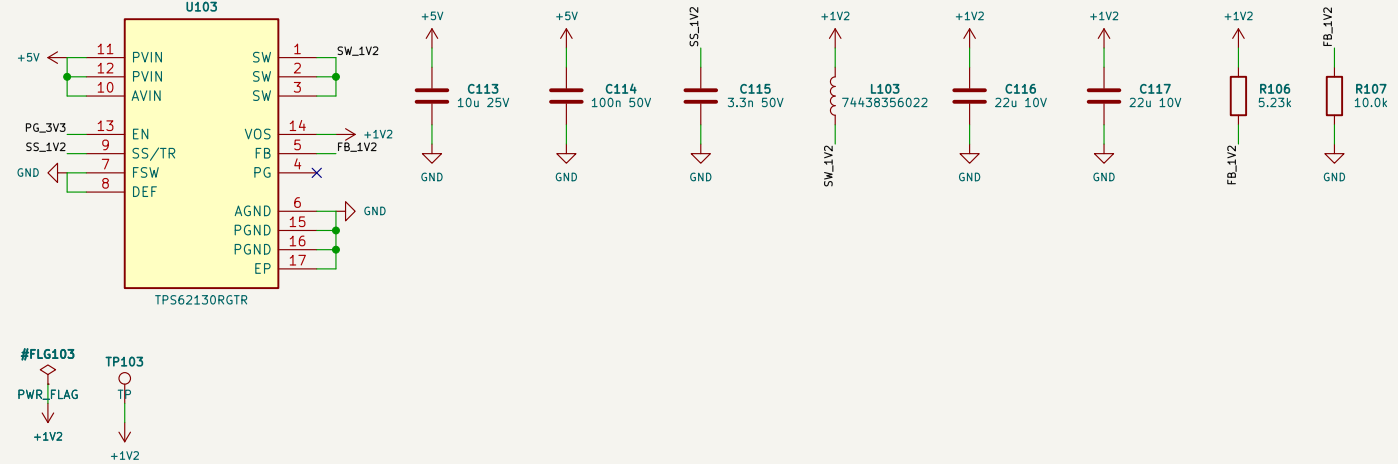
Table 9-1 row 400 kHz / 5 V / 2 A: L 15 uH, 3x22 uF, RFBT 100k / RFBB 24.9k -> 5.02 V, RT 64.9k.
Load (calculated): 3.3 V 1.3 A + 1.2 V 0.48 A + 3.3 V analog 0.1 A -> about 1.2 A at 5 V.
EN tied to VIN (datasheet allows), PG_5V enables the 3.3 V converter.
Output capacitance check (SNVSC40E, calculated): total on +5V = 3x22 + 2x10 + 1 + 0.3 = 87 uF nominal,
96 uF at +10 %; 1.45x the 66 uF design value, inside the 9.2,2.5 limit (10x design or 1000 uF) -> no extra
stability study required. Start-up: tSS 4.0 ms (8.3.9) -> 96 uF x 5.02 V / 4 ms = 0.12 A charge current,
downstream loads held off by the PG chain, << ILS-LIMIT 1.8 A min; OCP blanking 18 ms -> no hiccup.
Inductor: LMIN = 0.25 x 5 V / 400 kHz = 3.1 uH (eq. 11) < 15 uH; ripple 0.64-0.68 A at 21.6-26.4 V in
(> 10 % of 2 A); peak 1.54 A (1.62 A at L -20 %) < IHS-LIMIT 2.6 A min < ISAT 5.55 A. Bode not measured.

5 V -> 3.3 V DIGITAL (TPS62133, 3 A)



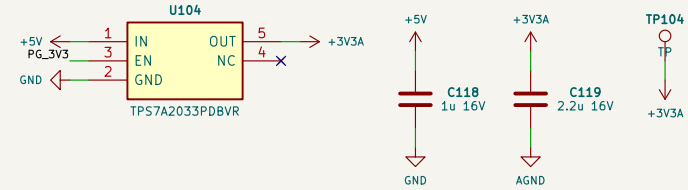
Fixed 3.3 V; FB to AGND per datasheet, FSW low (2.5 MHz), 2.2 uH, 2x22 uF (table 9-1).
SS 3.3 nF (typical application), PG_3V3 releases the 1.2 V and 3.3 V analog rails.
Worst case 3.221-3.396 V (power-save mode, SLVSAG7F p5) inside VDDIO 3.14-3.47 V.

5 V -> 1.2 V CORE (TPS62130, 3 A)



VOUT = 0.8 V x (1 + 5.23k/10.0k) = 1.218 V (eq. 6), 0.1 % 10 ppm/K; worst case 1.1872-1.2558 V
(power-save VFB 781.6-822.4 mV, FB leakage, 0.1 % line/load) inside VDD 1.14-1.26 V (CSR-01).
Enabled by PG_3V3: VDD rises after VDDIO (SPRSP14E table 7-1). VDD bulk on the decoupling sheet.

5 V -> 3.3 V ANALOG (TPS7A2033)



VDDA, op amps, reference, mux. Enabled by PG_3V3 so VDDA never leads VDDIO.
Load (calculated): 26 op-amp channels x 2.6 mA max + IDDA 15 mA + REF and its 4.2 mA load + VBIAS
+ 12 open-input bias resistors to +3V3A (16 mA) = about 110 mA of 300 mA.

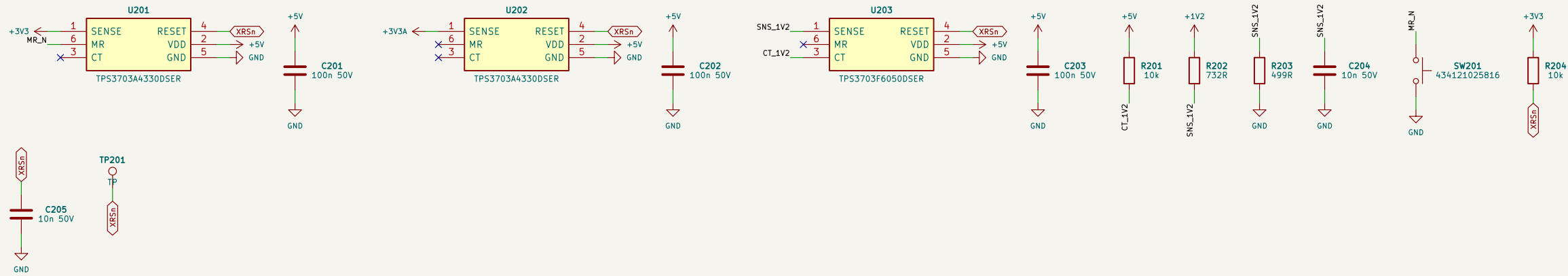
Not bench-validated. Values marked calculated are not simulated.
Sheet 2
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /01 Power regulators/
File: 01_power.kicad_sch

Title: CTRL-C2000 control card - Power regulators

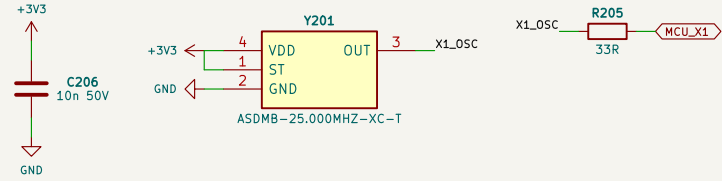
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 2/15

SUPPLY SUPERVISORS (OPEN-DRAIN RESETS WIRED TO XRSN), CSR-01



SPRSP14E 7.10.1.2.4: VDDIO and VDD need an external supervisor (the internal POR trips below range). Worst case at every corner (asserted on each build, MCU window 3.14–3.47 V / 1.14–1.26 V):
+3V3 A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2209$; OV 3.408–3.456 $> \text{regulator max } 3.3958$ V
+3V3A A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2505$; OV 3.408–3.456 $> \text{regulator max } 3.3495$ V
+1V2 F6050: trips ≥ 1.1454 , releases $\leq 1.1829 < \text{regulator min } 1.1872$; regulator max 1.2558 V
1.2 V is UV-only: a $\pm 4\%$ window cannot hold the TPS62130 power-save spread; its maximum is the regulator arithmetic above. Delay: CT open 10 ms (A parts), CT to VDD via 10k 20 ms (F part). VDD = +5V so the outputs stay defined while 3.3 V collapses. XRSn pull-up 10k, 10 nF; SYS_RST_N and the JTAG RESET are open drain on the same node. Reset button on MR of the 3.3 V part (100k internal pull-up).

25 MHZ CLOCK (X1)



SPRSP14E 7.10.3.2.1: X1 from an oscillator 10–25 MHz, tr/ff ≤ 6 ns, 45–55 %; X2 open (figure 7–11). No crystal, so VSSOSC goes to board ground (table 6–12). 33 R source termination.

BOOT MODE (GPIO72 = PIN 1, GPIO84 = PIN 0)

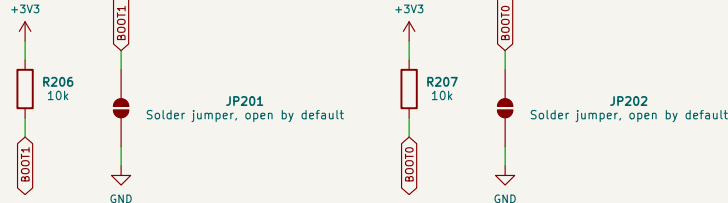
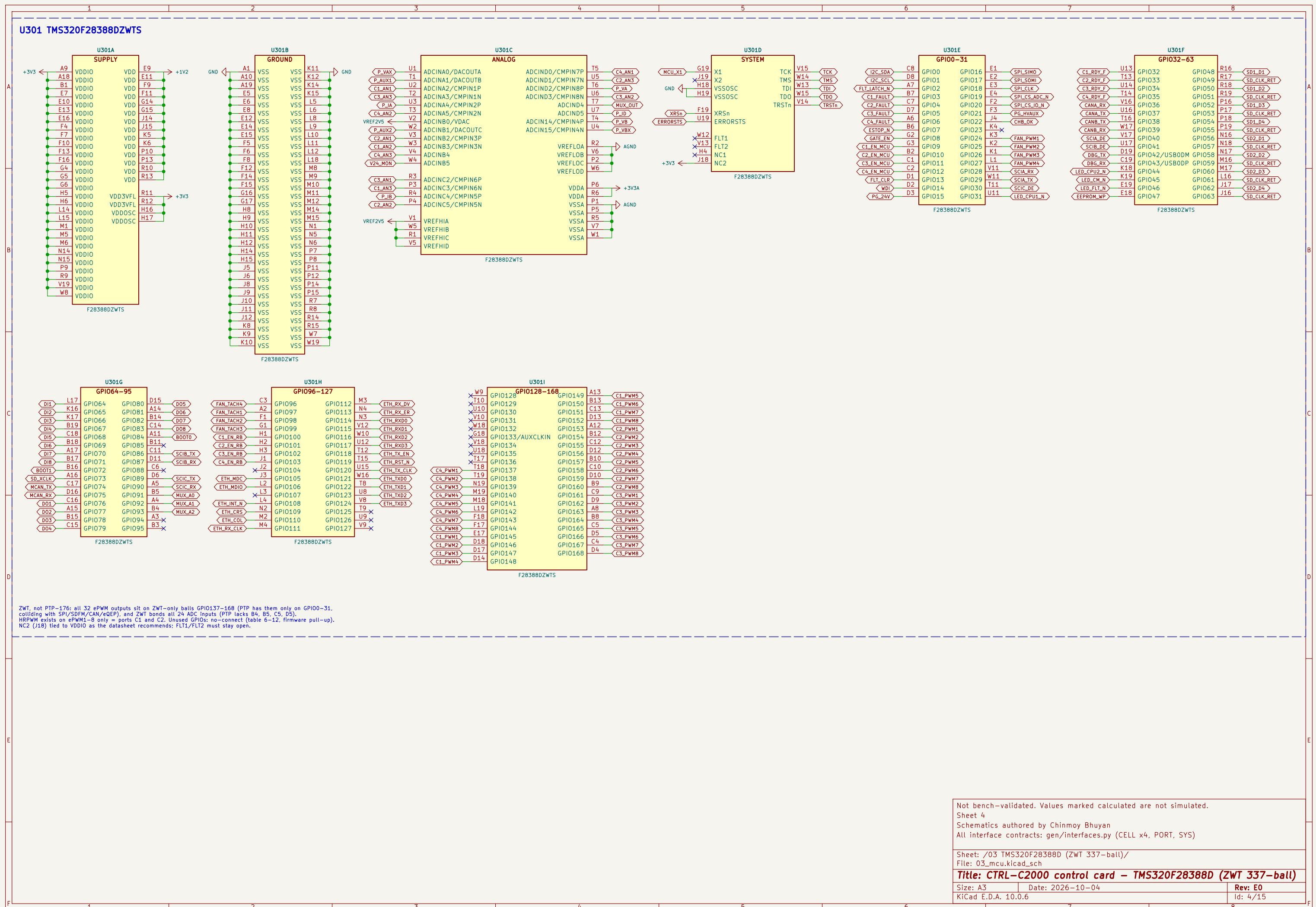
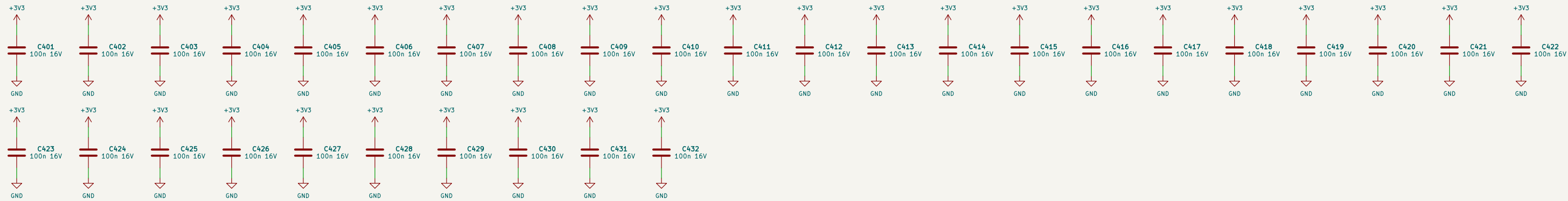


Table 8–18: 11 = flash (straps open), 01 = SCI, 10 = CAN, 00 = parallel I/O. Close JP to pull a pin low. th(boot-mode) 1.5 ms; nothing else drives these balls.

Not bench-validated. Values marked calculated are not simulated.
Sheet 3
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

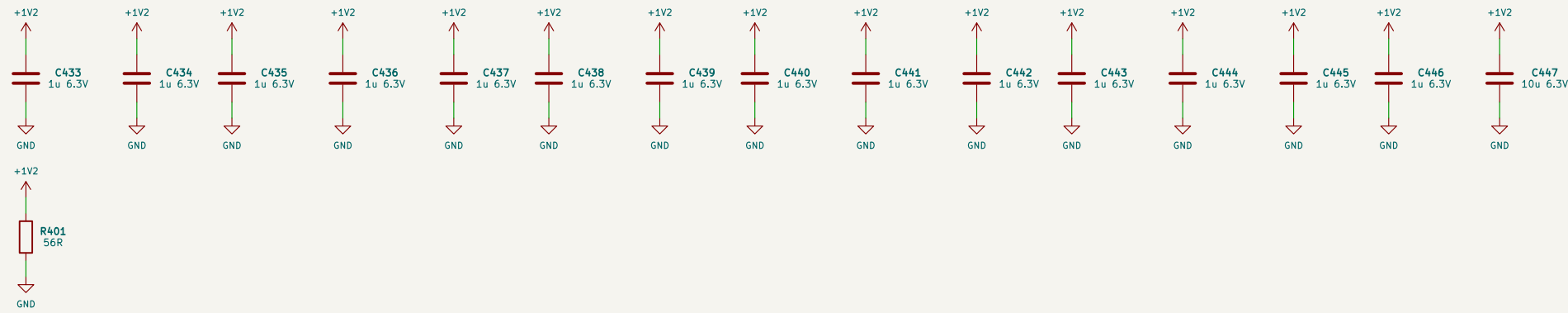


VDDIO (28 BALLS) + VDD3VFL (2) + VDDOSC (2): 0.1 UF EACH



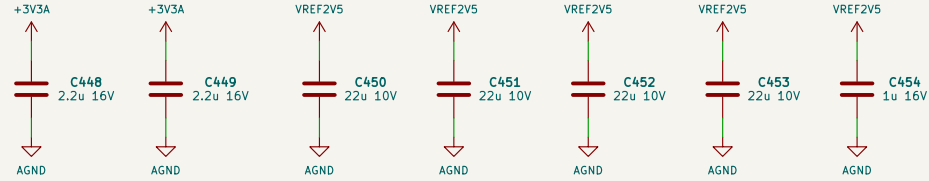
Configuration 1 of 7:10.1.3.2.

VDD 1.2 V (14 BALLS)



14 x 1 uF + 10 uF = 24 uF, 10 % parts: >= 21.6 uF (CVDD TOTAL 20 min / 22 typ. max tolerance 20 %).
56 R load consumes the VDD3VFL-to-VDD leakage current (table 6-4 note).

VDDA, VREFHI AND VDAC



VDDA 2.2 uF per ball to VSSA. VREFHIA-D: 22 uF each to VREFLO (16-bit mode).
VDAC (ADCINB0, V2) = CMPSS DAC reference: >= 1 uF at the ball (table 6-1).
All on VREF2V5: 4 x 22 + 1 + 1 uF = 90 uF, 99 uF at +10 %: inside the REF5025E 1-100 uF range (summed from the design on every build).

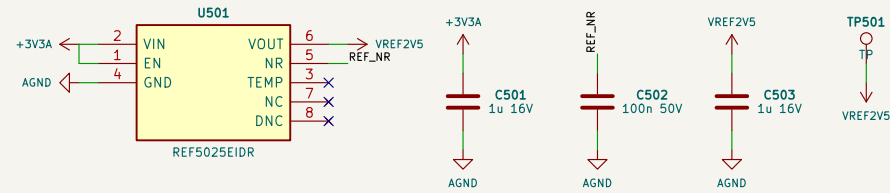
Not bench-validated. Values marked calculated are not simulated.
Sheet 5
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /04 MCU supply decoupling/
File: 04_decoupling.kicad_sch

Title: CTRL-C2000 control card - MCU supply decoupling

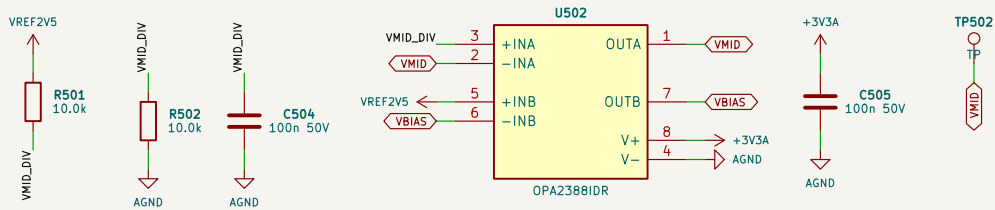
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 5/15

2.5 V ADC AND CMPSS DAC REFERENCE



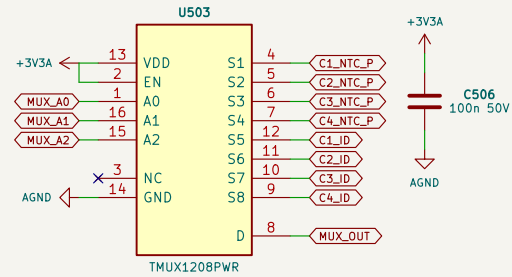
Powered from +3V3A (VIN $\geq$ VOUT + 0.2 V) so VREFHI and VDAC can never exceed VDDA (SPRSP14E 7.11.2.3.5, 7.11.3.1.4). Also the CMPSS VDAC reference (VDAC, INT=05): thresholds ratimetric with the ADC. Load 4.2 mA of +/-10 mA: DAC 8 x 6 k (3.3 mA) + VREFHI 4 x 190 uA + VMID divider (asserted). NR 0.1 uF class 1; 1 uF HF at VOUT; VREFHI and VDAC capacitors are on the decoupling sheet.

VMID AND VBIAS BUFFERS



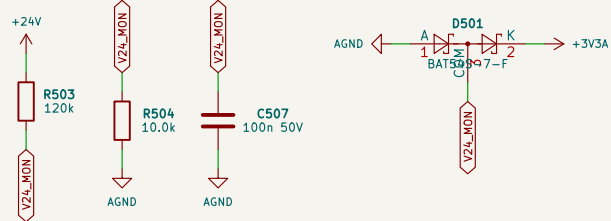
VMID = VREF/2 is ratiometric: a zero differential input reads mid-scale whatever the reference error.
VBIAS = VREF buffered, so ID and NTC readings are ratiometric and the reference itself is not loaded.

SLOW-SIGNAL MUX -> ADCIND4



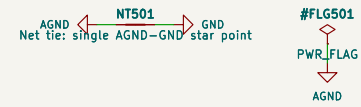
S1-S4 = C1-C4 NTC, S5-S8 = C1-C4 ID (24 ADC balls cannot take 20 fast channels + 9 slow ones). Each input has its own 100 nF on the cell sheet; the mux only switches between charged caps.

24 V MONITOR -> ADCINB5



120k/10k: 24 V reads 1.85 V, full scale 32.5 V; 100 nF filter.
+24V is live before VDDA: the BAT54S holds the ball at VDDA + VF (7.10.1.4.2).

AGND – GND STAR POINT



Single connection of the analog return to board ground (layout: under U301).

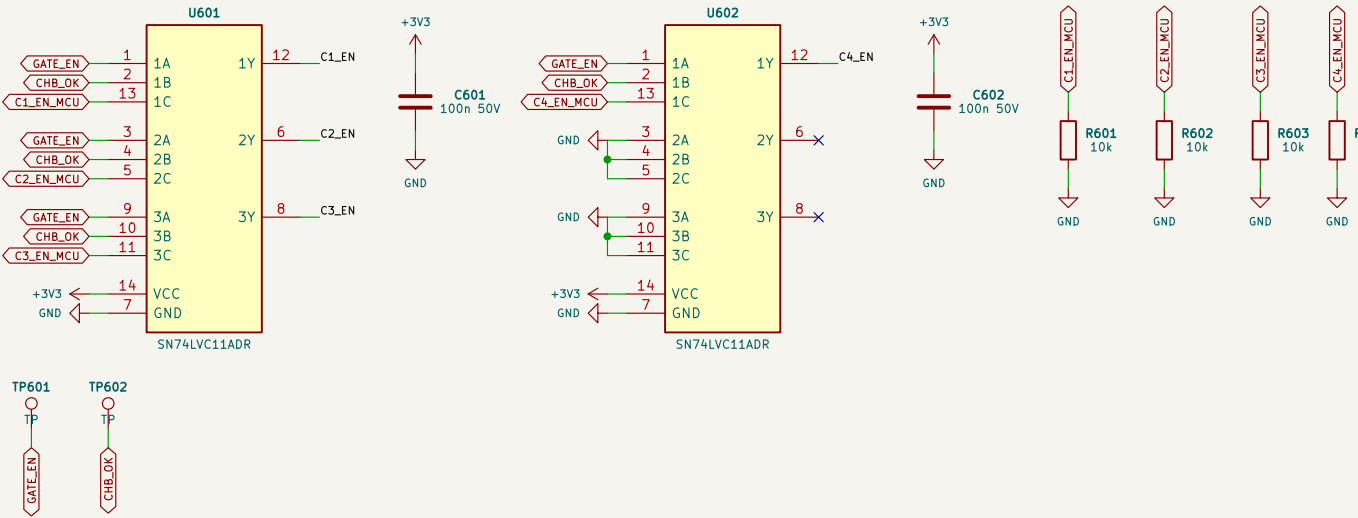
Not bench-validated. Values marked calculated are not simulated.
Sheet 6
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /05 ADC reference, VMID, slow mux/
File: 05_adcref.kicad_sch

Title: CTRL-C2000 control card – ADC reference, VMID, slow mux

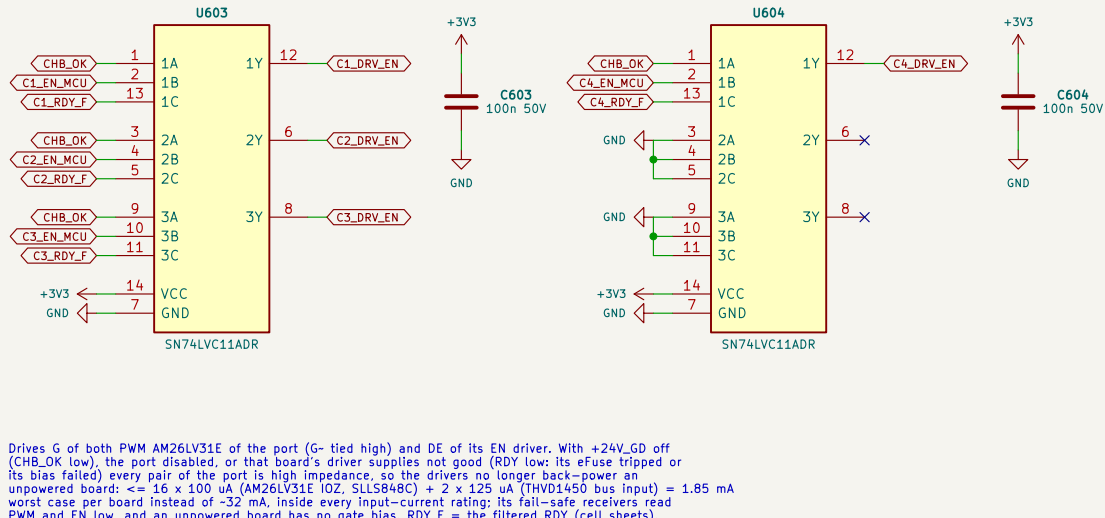
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 6/15

HARDWARE GATE-ENABLE AND (BOTH SAFETY CHANNELS)



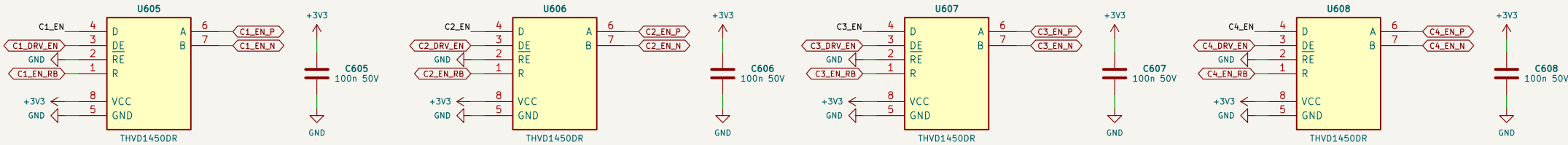
Cn_EN = GATE_EN (channel A) AND CHB_OK (channel B) AND Cn_EN_MCU, one 3-input gate per port; firmware alone can never raise an EN pair. Channel B also removes +24V_GD (gate-driver bias) on SYS. 10k pull-downs hold each MCU enable low in reset; GATE_EN and CHB_OK are pulled low on the SYS side. Spare gates: inputs to GND, outputs open.

LINE-DRIVER ENABLE CN_DRV_EN = CHB_OK AND CN_EN_MCU AND CN_RDY (INT-15)



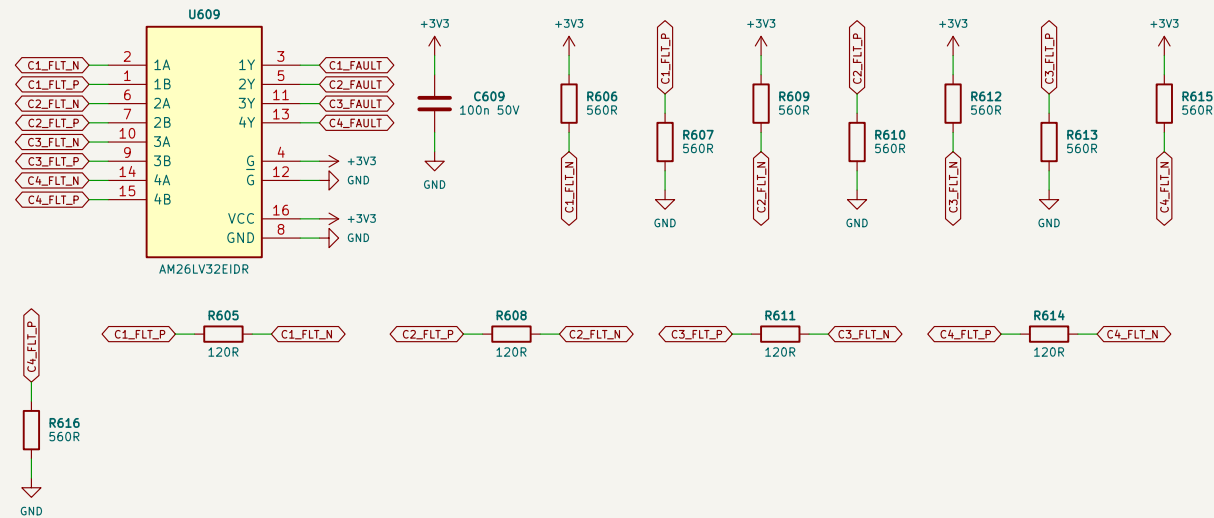
Drives G of both PWM AM26LV31E of the port (G- tied high) and DE of its EN driver. With +24V_GD off (CHB_OK low), the port disabled; or that board's driver supplies not good (RDY low: its eFuse tripped or its bias failed) every pair of the port is high impedance, so the drivers no longer back-power an unpowered board: $\leq 16 \times 100 \mu\text{A}$ (AM26LV31E IOZ, SLLS848C) + $2 \times 125 \mu\text{A}$ (THVD1450 bus input) = 1.85 mA worst case per board instead of -32 mA, inside every input-current rating: its fail-safe receivers read PWM and EN low, and an unpowered board has no gate bias. RDY_F = the filtered RDY (cell sheets).

EN LINE DRIVERS WITH READBACK (C1-C4)



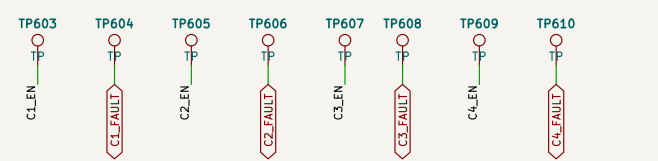
THVD1450 per port: D = Cn_EN, DE = Cn_DRV_EN, A/B = EN_P/EN_N, [VOD] $\geq 2 \text{ V}$ into 100 R (SLLSEY3E 7.7). Receiver always on (RE- low): R = the pair's state at the CTRL end $\rightarrow$ Cn_EN_RB on a spare GPIO, so a stuck AND gate or driver is visible (INT-22: start-up and proof-test sequence in the pin plan).

FLT RECEIVERS (C1-C4), FAIL-SAFE



Pair wired swapped (A = FLT_N, B = FLT_P): Cn_FAULT is high = fault. 560R bias + 120R termination put +0.32 V across an open/unplugged pair ($\rightarrow$ VIT+ 0.2 V) $\rightarrow$ fault; the AM26LV32E rev E only has open-circuit (not terminated) fail-safe, hence the external bias. Healthy: driver -2 V $\rightarrow$ low, Cn_FAULT $\rightarrow$ INPUTXBAR2-5 $\rightarrow$ ePWM X-BAR $\rightarrow$ TRIP4/5/7/8 (no firmware in the trip path).

TEST POINTS



Gated EN (after the AND) and receiver FAULT outputs, one pair per port.

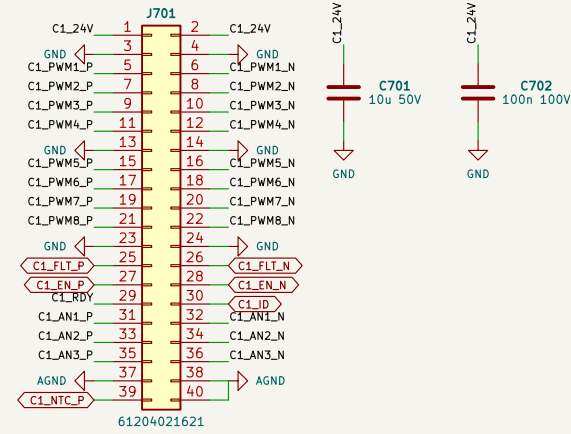
Not bench-validated. Values marked calculated are not simulated.
Sheet 7
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /06 Fault receivers, enable, EN drivers/
File: 06_trip.kicad_sch

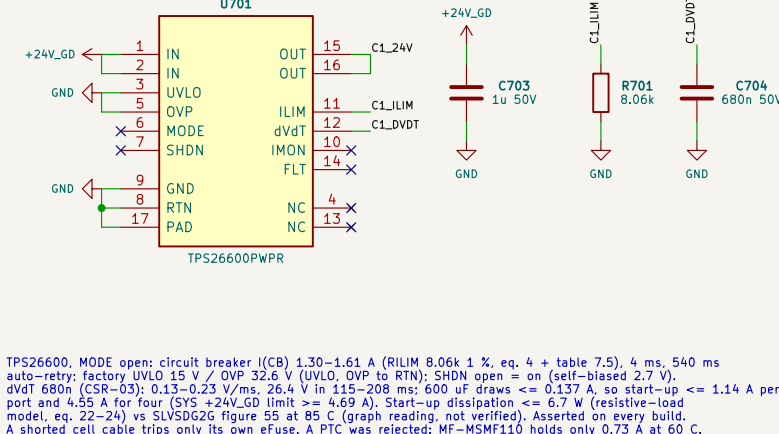
Title: CTRL-C2000 control card - Fault receivers, enable, EN drivers

Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 7/15

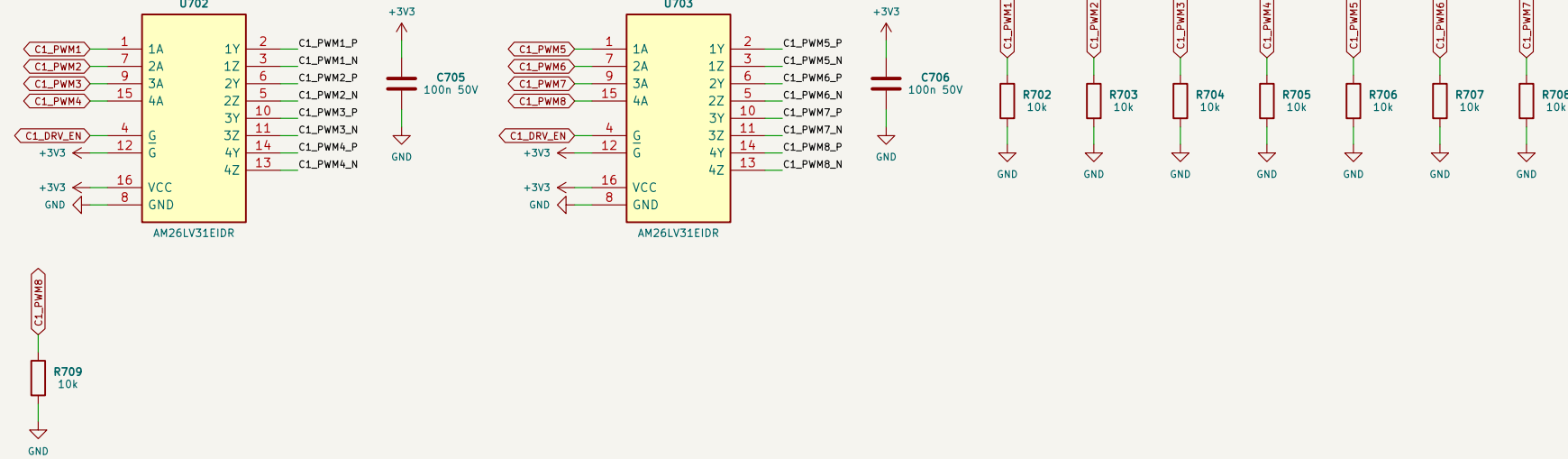
C1 CONNECTOR (CELL CONTRACT)



+24V_GD EFUSE FOR C1 (<= 1.0 A CONTINUOUS, <= 600 uF)

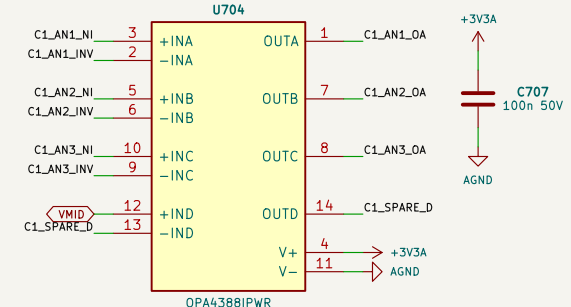


PWM LINE DRIVERS (EPWM1-4)



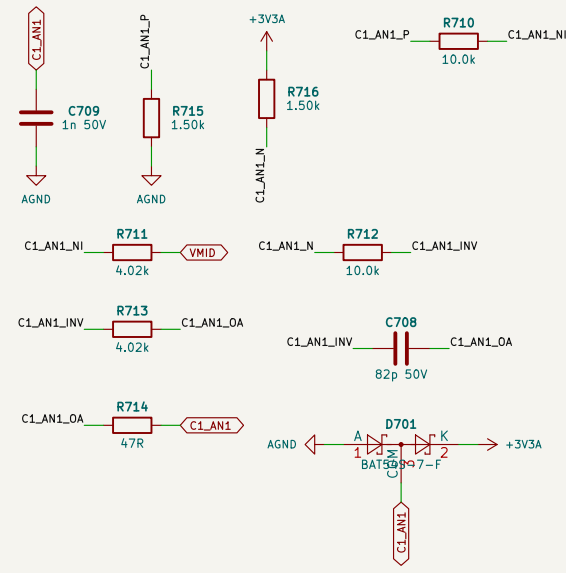
PWMx (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = CL_DRV_EN, G=high: outputs high impedance unless CHB_OK, CL_EN_MCU and RDY are high (N1-15). ePWM1-4 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

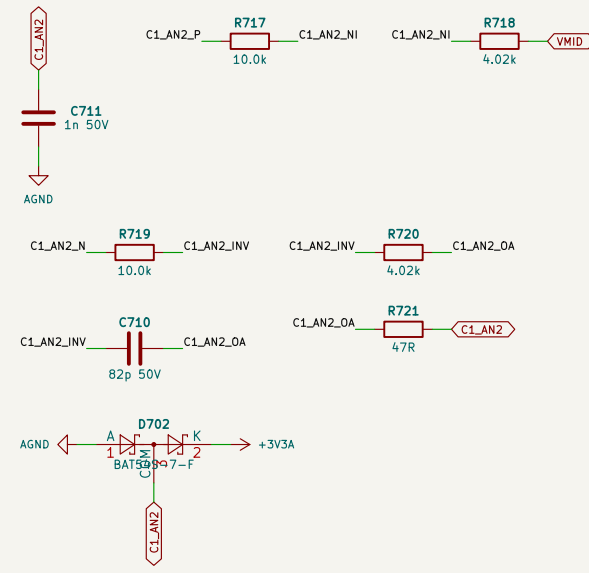


Difference amp per pair: $V_{ad} = V_{MD} 1.25 V + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$. C1 || Rf anti-alias pole 483 kHz, 47 R + 1 nF CG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT5NS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated. Accepts on any AN pair: differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with AN_N = AGND at the source (temperatures) -> 1.25-2.46 V, Op-amp inputs 0.9-1.75 V (RMID). Source resistance adds to the 10.0k: keep R <= 100 R (1 % gain) or calibrate it out. AN1 = inductor current (INT-12, one constant); PVCELL-25 IEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11k A -> 8.0k mV/A at the ADC. 0.0759 A/V/LSB; CMPS backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it. DAB-360 (C1/C2): AN1 5.009 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMD follower.

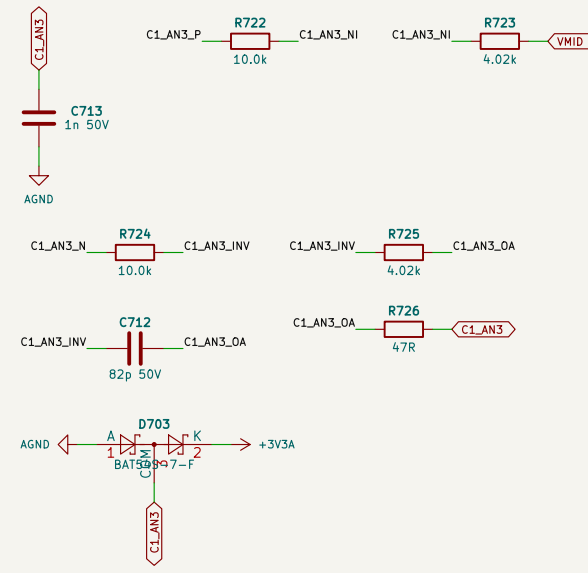
C1_AN1 -> ADCINA2 / CMPIN1P



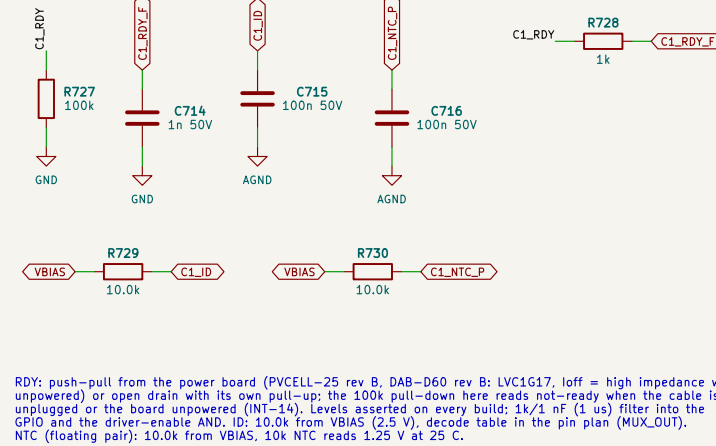
C1_AN2 -> ADCINB3



C1_AN3 -> ADCINC3



RDY, ID, NTC



RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (N1-14). Levels asserted on every build; 1k/2 nF (1 us) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 8
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /07 Cell port C1/
File: 07_cell1.kicad_sch
Title: CTRL-C2000 control card - Cell port C1

Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 8/15

Pinout diagram for the 40-pin J801 connector. The diagram shows two rows of pins. The top row (pins 1-20) is labeled C2_24V, GND, C2_PWM1_P, C2_PWM2_P, C2_PWM3_P, C2_PWM4_P, GND, C2_PWM5_P, C2_PWM6_P, C2_PWM7_P, C2_PWM8_P, C2_FILT_P, C2_LED_P, C2_RDY_P, C2_AN1_P, C2_AN2_P, C2_AN3_P, AGND, and C2_NTC_P. The bottom row (pins 21-40) is labeled C2_24V, GND, C2_PWM1_N, C2_PWM2_N, C2_PWM3_N, C2_PWM4_N, GND, C2_PWM5_N, C2_PWM6_N, C2_PWM7_N, C2_PWM8_N, GND, C2_FILT_N, C2_LED_N, C2_ID, C2_AIN1_N, C2_AIN2_N, C2_AIN3_N, AGND, and C2_NTC_N. Two capacitors are shown: C801 (10u 50V) connected between pins 2 and 3, and C802 (100n 100V) connected between pins 21 and 22.

NTC_N returns to AGND here; PCT/EN pairs go to the chip sheet.

[illegible][illegible]

PWMk (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs: ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = C2_DRV_EN, G_high: outputs high impedance unless CHB_OK, C2_PWM_MCU and RDI are high (INT-15). ePWM5-8 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

OPA4388IPWR

Pin	Function	Connection
1	INA	C2_AN1_DA
2	INB	C2_AN2_DA
3	INB	C2_AN2_INV
4	INB	C2_AN2_INV
5	INB	C2_AN2_INV
6	INB	C2_AN2_INV
7	OUTB	C2_AN2_DA
8	OUTC	C2_AN3_DA
9	INB	C2_AN3_INV
10	INB	C2_AN3_INV
11	INB	C2_AN3_INV
12	INB	C2_AN3_INV
13	INB	C2_AN3_INV
14	OUTD	C2_SPARE_D
15	V+	+3V3
16	V-	AGND

OPA4388IPWR

Difference amp per pair: V_{DS} = 125 V; V_{GS} = 6 G (ANP - ANLN); IR_{ON} 10.0 ± 0.1 mS/km; R_{DS(on)} = 6.42 Ω/10.0 kΩ × 0.402 C_{iss} [fF]; R_f anti-alias probe 483 kΩ; I_{DS} = 1 nFC; ADC charge bucket (ACQPS = 30 pmV×ns at 12 bit). BAT54S channels the ADC node to 3V3/AGND. All values calculated, not simulated.

Output swing range: V_{OUT} = 0.75 V (ANP) + 0.75 V (ANLN) = 1.5 V (V_{DC} snap +3.1 V), or 0–3.0 V single-ended with ANLN = AGND at the source (temperatures) → 1.25–2.46 V. Op-amp inputs over 0.175 V (RIN), Source resistance adds up to the 10.0k; keep it <= 100 R (1 g gain) or calculate it. A 100 MΩ load will add ~1.14 μA → 8.04 mV at the ADC, 0.0759 /4.18V; CMPSB bucking +91.5% The LA U150 P offset (up to 1.5 A) is re-zeroed by firmware at every idle (gate) off; trip bands carry only the 1.17 A left after it. 0.048 A @ 1.5 A is 0.032 V @ 1.58 V/mV; 0.048 A @ 1.5 A is 0.032 V @ 1.58 V/mV. Single-ended, 49.9 R sources (+0.5 X gain). Only ANL carries the open-input bias. Channel D: VMID follower.

[illegible][illegible][illegible]

Figure 1 is a schematic diagram of the power plane layout for the R827, R828, R829, and R830 components. The diagram shows the power and ground connections for each component, including the C2_RDY, C2_ID, C2_NTC_P, and C2_NTC_F capacitors. The components are connected to a common ground plane (GND) and a power plane (VBias). The R827 and R828 components are connected to GND and C2_RDY. The R829 component is connected to GND and C2_ID. The R830 component is connected to GND and C2_NTC_P. The C2_NTC_F capacitor is connected to C2_RDY and C2_NTC_P. The components are connected to a common ground plane (GND) and a power plane (VBias). The R827 and R828 components are connected to GND and C2_RDY. The R829 component is connected to GND and C2_ID. The R830 component is connected to GND and C2_NTC_P. The C2_NTC_F capacitor is connected to C2_RDY and C2_NTC_P.

R827 100k
GND
C2_RDY
C2_ID
C2_NTC_P
C2_NTC_F
R828 1k
C2_RDY_F
R829 10.0k
VBias
C2_ID
R830 10.0k
VBias
C2_NTC_P

Figure 1: Power plane layout for the R827, R828, R829, and R830 components.

R827: push-pull from the power board (PVCELL-25 rev B, DAB-D60 rev B; LVC1A17; I_{OH} = high impedance w/ unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered. (NT-14). Levels asserted on every valid 3V(1) NF (1 μ s) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBias; 10k NTC reads -2.5 V at 25 $^{\circ}$ C.

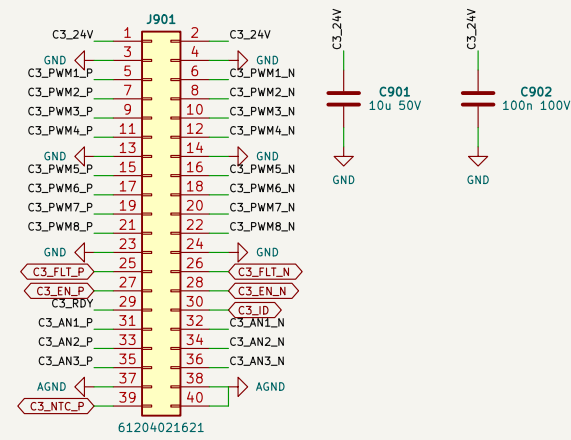
R828: push-pull from the power board (PVCELL-25 rev B, DAB-D60 rev B; LVC1A17; I_{OH} = high impedance w/ unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered. (NT-14). Levels asserted on every valid 3V(1) NF (1 μ s) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBias; 10k NTC reads -2.5 V at 25 $^{\circ}$ C.

R829: push-pull from the power board (PVCELL-25 rev B, DAB-D60 rev B; LVC1A17; I_{OH} = high impedance w/ unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered. (NT-14). Levels asserted on every valid 3V(1) NF (1 μ s) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBias; 10k NTC reads -2.5 V at 25 $^{\circ}$ C.

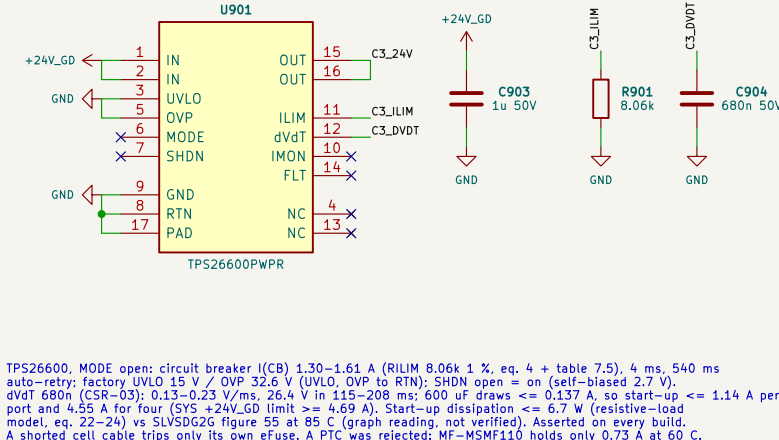
R830: push-pull from the power board (PVCELL-25 rev B, DAB-D60 rev B; LVC1A17; I_{OH} = high impedance w/ unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered. (NT-14). Levels asserted on every valid 3V(1) NF (1 μ s) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBias; 10k NTC reads -2.5 V at 25 $^{\circ}$ C.

RDY: push-pull from the power board (PVCCELL-25 rev B, DAB-D60 rev B; LVC1G17, Ioff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build; 1k/1 nF (1- μ s) filter into the GPIO and the driver-enable AND. ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pin): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

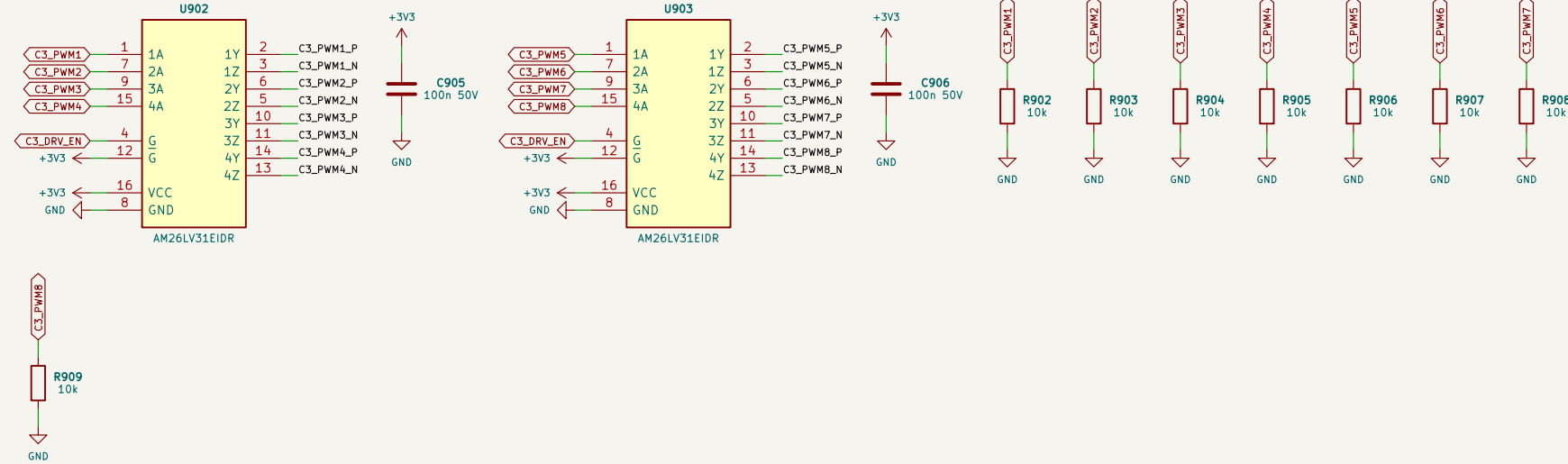
C3 CONNECTOR (CELL CONTRACT)



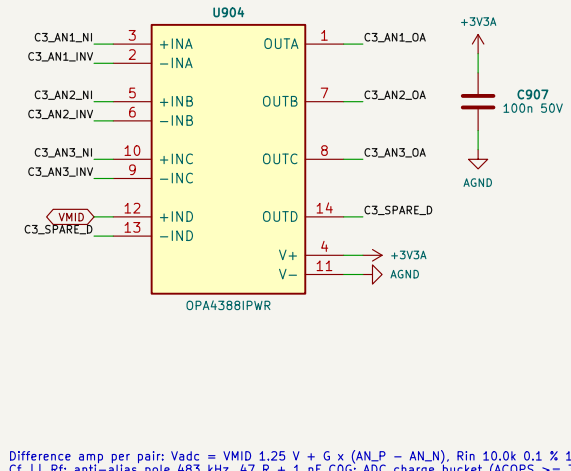
+24V_GD EFUSE FOR C3 (<= 1.0 A CONTINUOUS, <= 600 uF)



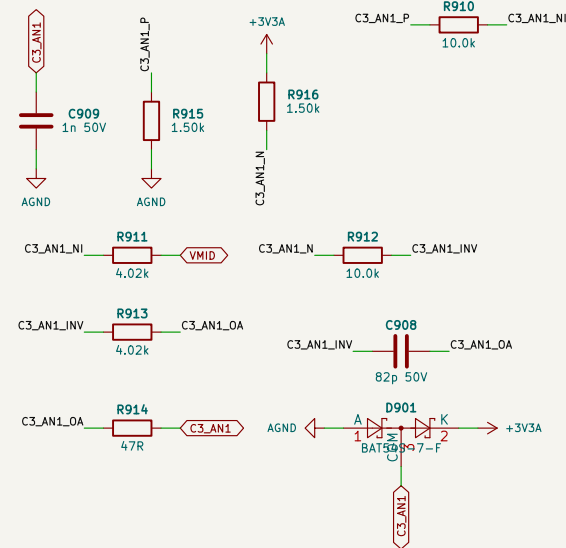
PWM LINE DRIVERS (EPWM9-12)



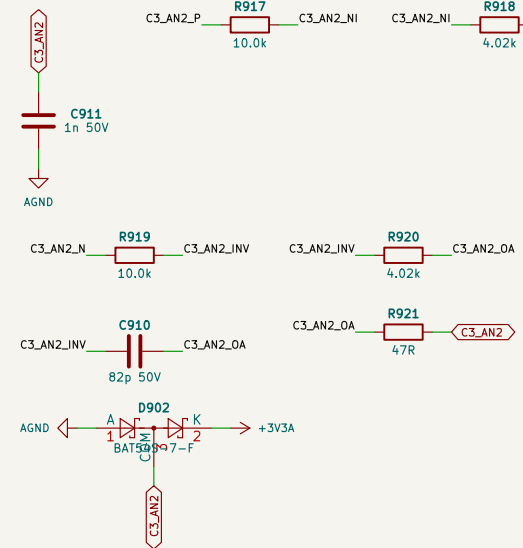
ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE



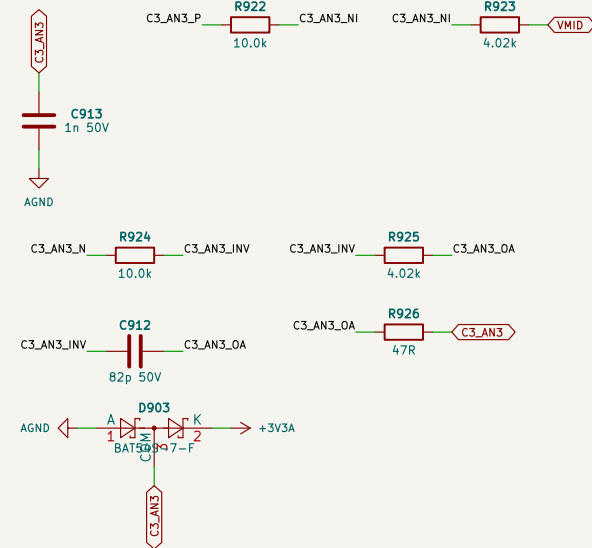
C3_AN1 -> ADCINC2 / CMPIN6P



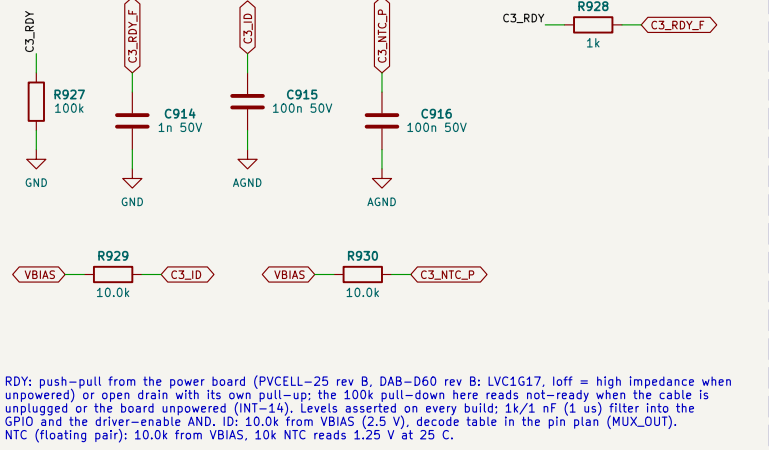
C3_AN2 -> ADCIND3



C3_AN3 -> ADCINA3

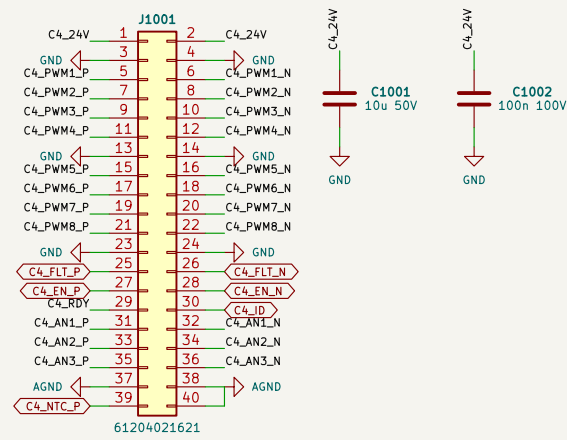


RDY, ID, NTC



Not bench-validated. Values marked calculated are not simulated.
Sheet 10
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

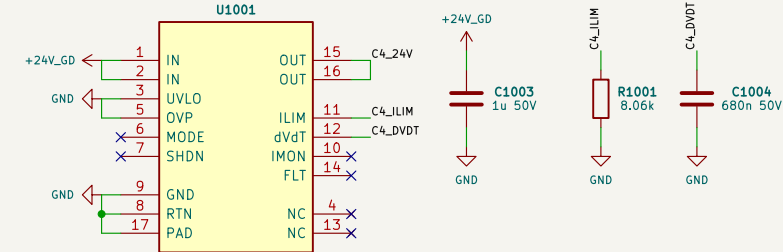
C4 CONNECTOR (CELL CONTRACT)



Pin map = Interfaces.CELL. Pins 1-2 carry +24V_GD only (gate-driver bias, channel B switched).

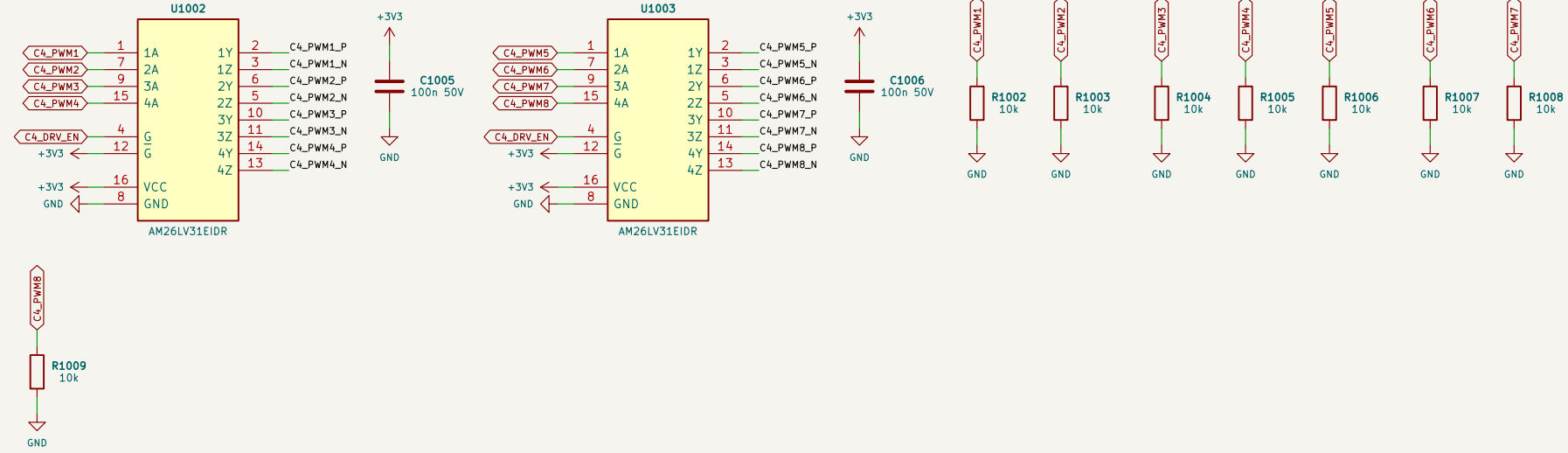
NTC_N returns to AGND here. FLV/EN gains go to the trip sheet.

+24V_GD EFUSE FOR C4 (<= 1.0 A CONTINUOUS, <= 600 uF)



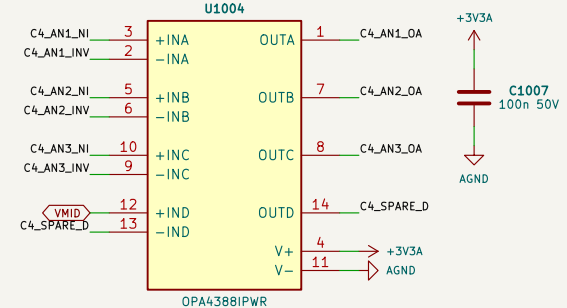
TPS26600, MODE open: circuit breaker I(CB) 1.30-1.61 A (RILIM 8.06k 1 %, eq. 4 + table 7.5), 4 ms, 540 ms auto-retry, factory UVLO 15 V / OVP 32.6 V (UVLO, OVP to RTN); SHDN open = on (self-biased 2.7 V), dVdI 680k (CSR-D3), 0.13-0.23 V/ms, 26.4 V in 115-208 ms; 600 uF draws <= 0.137 A, so start-up <= 1.14 A per port and <55 A for four (SYS <24V_GD limit <= 4.69 A). Start-up dissipation <= 6.7 W (resistive-load model, eq. 22-24) vs SLVSDG2G figure 55 at 85 C (graph reading, not verified). Asserted on every build. A shorted cell cable trips only its own eFuse. A PTC was rejected: MF-MSMF110 holds only 0.73 A at 60 C.

PWM LINE DRIVERS (EPWM13-16)



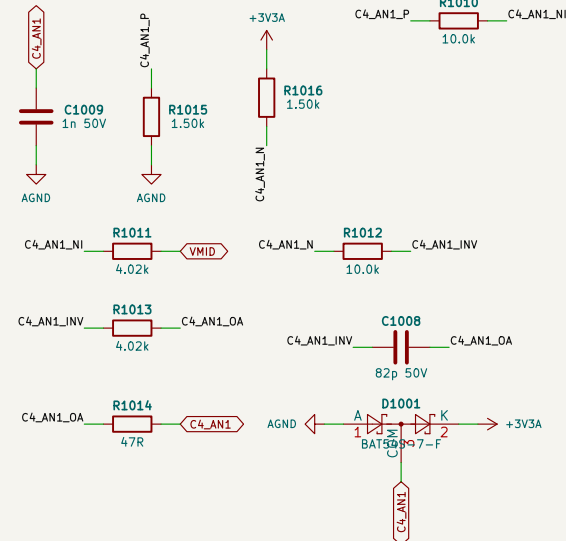
PWMs (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-8), so every gate command is 0 until firmware runs. G = C4_DRV_EN, G- high; outputs high impedance unless CHB_OK, C4_EN_MCU and RDY are high (INT-15). ePWM13-16 have no high-resolution PWM; buck-boost cells only, not a DAB bridge.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

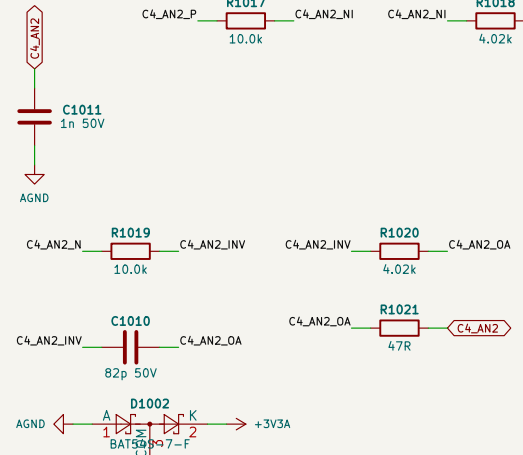


Difference amp per pair: $V_{\text{dcb}} = V_{\text{MID}} 1.25 \text{ V} + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$. Cf || Rf: anti-alias pole 483 kHz, 47 R + 1 nF COG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT5AS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated. Accepts on any AN pair: differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with ANL_N = AGND at the source (temperatures) -> 1.25-2.46 V. Op-amp inputs 0.9-1.75 V (RRIO). Source resistance adds to the 10.0k: keep R <= 100 R (1 % gain) or calibrate it out. AN1 = inductor current (INT-12, one constant); PVCELL-25 LEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11.4 A -> 8.0k mV/A at the ADC. 0.0759 A/LSD; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it. DAB-D60 (C1/C2): AN1 5.005 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMID follower.

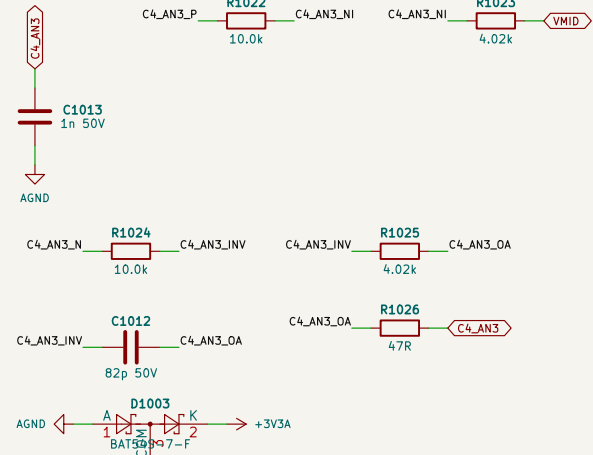
C4_AN1 -> ADCIND0 / CMPIN7P



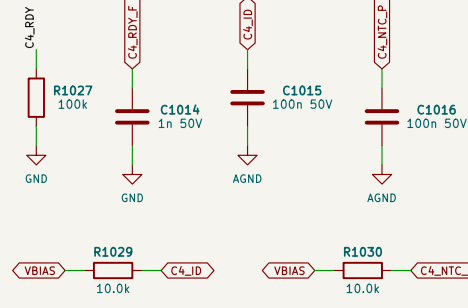
C4_AN2 -> ADCINA5



C4_AN3 -> ADCINB4



RDY, ID, NTC



RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build: 1k/1 nF (1 us) filter into the GPIO and the driver-enable AND, ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

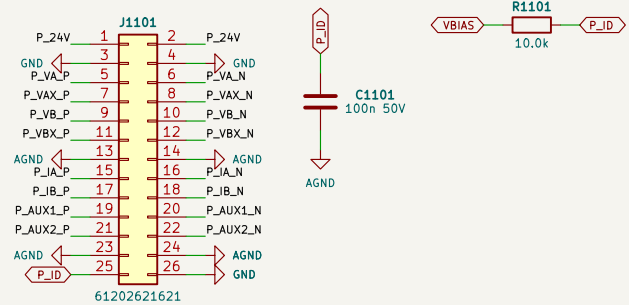
Not bench-validated. Values marked calculated are not simulated.
Sheet 11
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /10 Cell port C4/
File: 10_cell4.kicad_sch

Title: CTRL-C2000 control card - Cell port C4

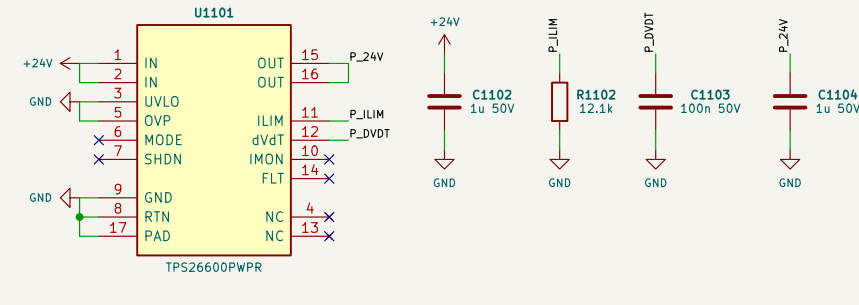
Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 11/15

PORT-BOARD CONNECTOR (PORT CONTRACT)



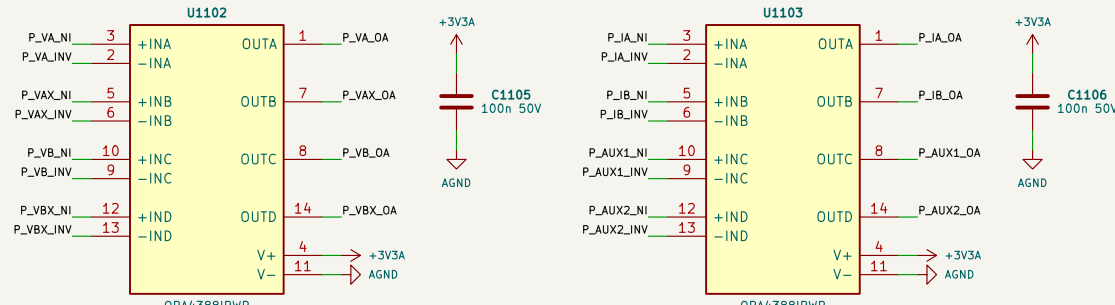
Pin map = Interfaces:PORT. P_ID: 10.0k from VBIAS. The port board sums its hold readbacks onto ID: decode table (both codes x 9 states, unpowered, open, short; every band >= 8 counts apart) in the pin plan. An unpowered board reads its own band inside the PV-PORT PPB window: firmware decodes it, and in hardware VA/VB read <= -161 V and trip (INT-07, asserted).

P_24V EFUSE (INT-06, CSR-12)



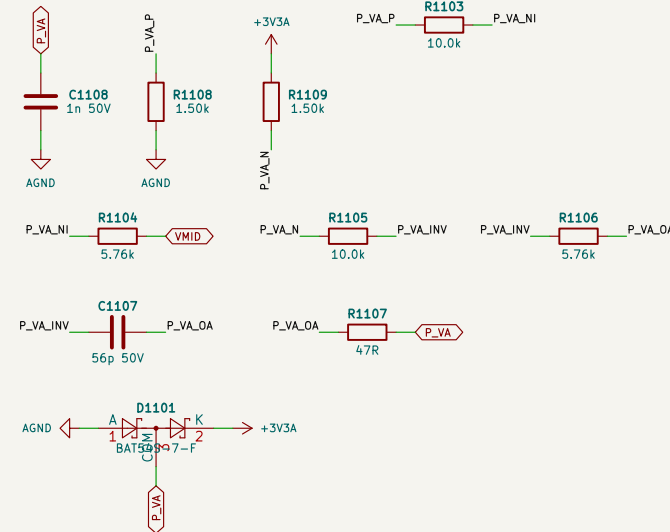
Port board: up to 0.7 A for up to 200 s (battery port: two hold-closed coils + sensors), TPS26600. MODE open: circuit breaker I(CB) 0.851-1.075 A (RILIM 12.1k 1 %), so the hold path never trips it; a shorted port board or cable trips only this branch (4 ms, 540 ms retry), and CTRL 0.35 A + 1.08 A stays below the SYS +24V eFuse (>= 1.82 A); the control card keeps running. dVdT 100n 0.86-1.56 V/ms into 5.3 uF. Factory UVLO 15 V / OVP 32.6 V; SHDN open. The PTC (0.73 A hold at 60 C) is gone.

ANALOG FRONT ENDS (2 X OPA4388)

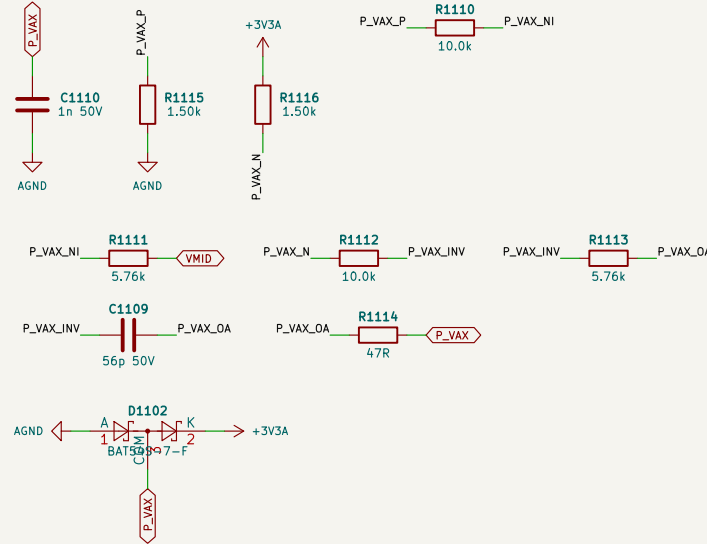


Difference amp per pair: $V_{adc} = V_{MID} 1.25 V + G \times (AN_P - AN_N)$, $R_{in} 10.0k 0.1 \% 10 ppm/K$, $G = 5.76k/10.0k = 0.576$; $\pm 2 V \rightarrow 0.10-2.40 V$. Cf [1] Rf anti-alias pole 493 kHz, 47 R + 1 nF CG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT54S clamps the ADC node to 3V3A/AGND. All values calculated, not simulated. Open-input bias (INT-07): 1.50k AN_P -> AGND and AN_N -> +3V3A, so an open conductor or ribbon reads far outside the trip window instead of 0 (worst readings: check trips build line). Loads the isolated amplifier with -1.3 k (AMC3330/AMC3302 RLOAD >= 1 k); source + cable resistance R gives a gain error $R/1.5 k (+ 0.07 \% \text{ for } 1 k)$, removed by the gain calibration. VA, VB, IB, IB sit on CMPS5 positive inputs: trip paths and bands in the pin plan.

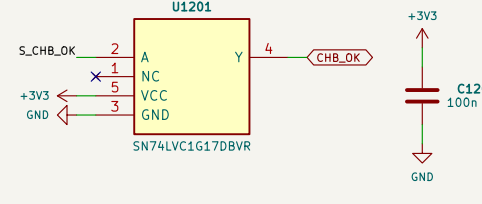
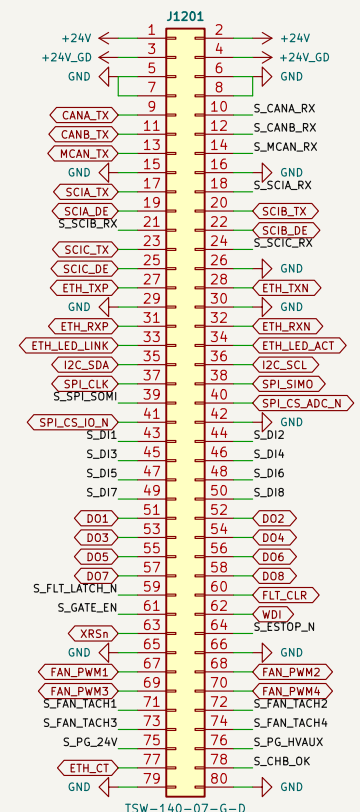
P_VA -> ADCIND2 / CMPIN8P



P_VAX -> ADCINA0



J1201			
+24V	1	2	+24V
+24V_GND	5	6	+24V_GND
GND	7	8	
CAMA_TX	9	10	S_CAMA_RX
CAN_B_TX	11	12	S_CANB_RX
MCAN_TX	13	14	S_MCAN_RX
CCO	15	16	OCN
SCIB_TX	17	18	S_SCIB_RX
SCIA_RX	19	20	SCIB_TX
SCIB_TX	21	22	SCIB_RX
SCIC_TX	23	24	S_SCIC_RX
ETH_TX	25	26	ETH_TX
GND	27	28	GND
GND	29	30	GND
ETH_RX2	31	32	ETH_RX2N
ETH_LED_LIN	33	34	ETH_LED_ACT
IO2_SDA	35	36	I2C_SCL
IO2_SCL	37	38	I2C_SDA
S_SPI_CS0N	39	40	S_SPI_CS0N
SPI_CS1_0N	41	42	GND
S_D01	43	44	S_D02
S_D03	45	46	S_D04
S_D05	47	48	S_D06
S_D07	49	50	S_D08
D01	51	52	D02
D03	53	54	D04
D05	55	56	D06
D07	57	58	D08
S_F1_L1_A0N	59	60	DIR
S_GATE1EN	61	62	CLK_CLKR
KRXS	63	64	S_ESTOP_N
FAN_PWM1	65	66	FAN_PWM2
FAN_PWM1	67	68	FAN_PWM2
FAN_TACH1	69	70	FAN_TACH2
S_FAN_TACH1	71	72	FAN_TACH2
S_FAN_TACH3	73	74	S_FAN_TACH4
S_PG_2V3	75	76	S_PG_1V8
ETH_C1	77	78	S_ETH_RX0
ETH_C2	79	80	GND

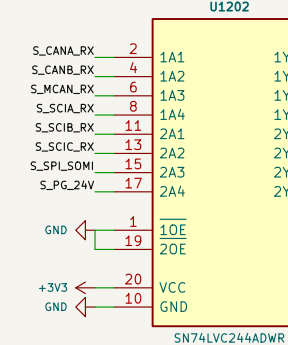


Schmitt buffer with 10k, separate from the 74LVC244A that carries GATE_EN, so one package cannot corrupt both safety channels. 10k pull-down: SYS unplugged = channel B not OK.

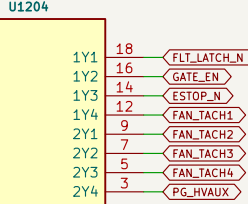
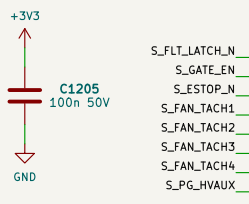
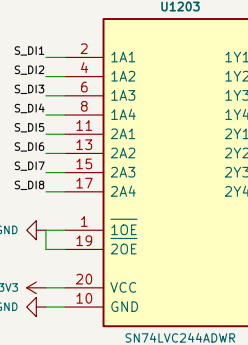
U1202

S_CANA_RX	2	1A1	1V
S_CANB_RX	4	1A2	1V
S_MCAN_RX	6	1A3	1V
S_SICIA_RX	8	1A4	1V
S_SICIB_RX	11	2A1	2V
S_SICIC_RX	13	2A2	2V
S_SPL_SOMI	15	2A3	2V
S_PG_24V	17	2A4	2V
GND	1	10E	20E
+3V3	20	VCC	
GND	19	GND	

SN74VLC244ADR

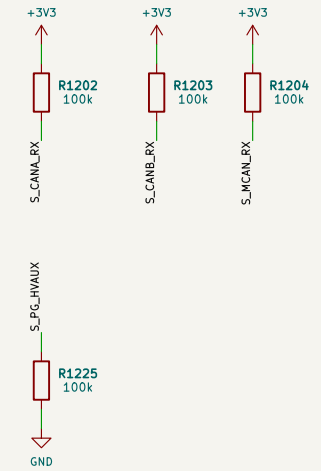


SYS makes its own 3.3 V and may drive these lines before this card's VDDIO is up; SPRSP14E 7.1 allows ± 20 mA total input-clamp current and 7.10.1.4.2 forbids driving pins early. 74LVC244A inputs and outputs are high impedance unpowered (loff) and its outputs never exceed VDDIO. tpd -5 ns.

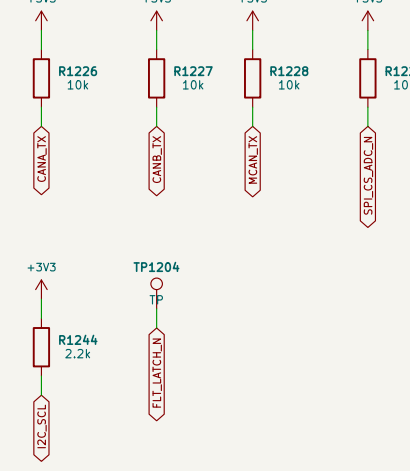


Pin map = interfaces.SYS: SYS_RST_N lands directly on XRSn (open drain both sides).
 +24V (pins 1-2): this card's regulators and the PORT connector. +24V_GD (pins 3-4): switched by safety channel. 18 on SYS: goes ONLY to the four cell eFuses (cell sheets). The two never meet here. ETH_CT (77): PHY analog supply to the magnetics centre taps (Ethernet sheet).
 Inputs from SYS arrive as `S_<name>` and reach the MCU as `<name>` through the buffers.

The schematic diagram shows the power supply section. It includes three resistors (R1202, R1203, R1204) connected in parallel between the +3V3 supply and the S_3ANL_RX signal line. Additionally, there is a resistor (R1225) connected between the S_3V3_LHUX signal line and the GND (ground) symbol.



FLT_LATCH_N, GATE_EN, ESTOP_N: 10k to GND, so an unplugged SYS reads as tripped and the gates stay off.
Other inputs: 100k to their idle level (CAN/SCI receive high, the rest low).

[illegible]

GPIOs float in reset: CAN TX recessive, RS-485 DE off, DO and FLT_CLR low, SPI chip selects high
I2C: 2.2k pull-ups here (bus master side); SYS should not add a second pair.

Not bench-validated. Values marked calculated are not simulated.
Sheet 13
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /12 SYS-IO-AUX connector/
File: 12_sys.kicad_sch

Title: CTRL-C2000 control card - SYS-IO-AUX connector

Size: A2

Date: 2026-10-04

Rev: E0

KiCad E.D.A. 10.0.6	Id: 13/15
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Diagram illustrating the JTAG pin connections for the FTSH-110-01-L-DV-K device. The connections are as follows:

- Pin 1:** TMS
- Pin 2:** TRStn
- Pin 3:** JTAG_VREF
- Pin 4:** GND
- Pin 5:** TDI
- Pin 6:** +3V3
- Pin 7:** JTAG_TDO
- Pin 8:** R1401 2.2k
- Pin 9:** TCK
- Pin 10:** +3V3
- Pin 11:** GND
- Pin 12:** R1402 2.2k
- Pin 13:** JTAG_EMU0
- Pin 14:** +3V3
- Pin 15:** XRSn
- Pin 16:** R1403 100R
- Pin 17:** GND
- Pin 18:** +3V3
- Pin 19:** GND
- Pin 20:** JTAG_EMU1

Additional components shown include resistors R1401 (2.2k), R1402 (2.2k), R1403 (100R), R1404 (4.7k), and R1405 (4.7k).

TMS 2.2k pull-up and TRSTn 2.2k pull-down (SPRSP14E table 6-1); TDIS to GND; VTRef via 100 R; RTCK looped to TCK; EMU0/1 4.7k pull-ups (no EMU pins on F2838x); RESET (open drain) on XRSn; EMU2-EMU4 (17-19) grounded as SPRSP14E 7.10.7 requires (CSR-07).

Pin 1 GND, 2 TX (from MCU), 3 RX (to MCU); 100 R series each way.

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Address 1010111 (A2-A0 high = 0x57) so it cannot collide with a SYS EEPROM at the default 0x50.
 WP pulled high: write-protected unless firmware drives EEPROM_WP low.

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WP pulled high: write-protected unless firmware drives EEPROM_WP low.

STATUS LEDs AND ERRORSTS

The diagram illustrates the circuit for STATUS LEDs and ERRORSTS. It features five LEDs (D1401-D1405) connected to +3V3 through resistors (R1409-R1413) and to GND. The LEDs are labeled LED_PWR_A, LED_CPU1_A, LED_CPU2_A, LED_CM_A, and LED_FLT_A. The ERRORSTS pin is connected to +3V3 through a 10k resistor (R1414) and to GND through a pull-down resistor (D1401).

GPIO sinks about 2 mA (680 R). ERRORSTS needs an external pull-down.

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SDFM EXPANSION HEADER (NO MATING CONNECT YET)

The diagram illustrates the SDFM Expansion Header (J1403) with the following connections:

- Power and Ground:**
 - Pin 1: +3V3
 - Pin 2: GND
 - Pin 17: +3V3
 - Pin 18: GND
- Clock Signals:**
 - Pin 3: SD_CLK_OUT
 - Pin 4: GND
 - Pin 5: SD_CLK_RET
 - Pin 6: GND
 - Pin 19: +3V3
 - Pin 20: GND
- Data Inputs (SD1 and SD2):**
 - Pin 7: SD1_D1
 - Pin 8: SD1_D2
 - Pin 9: SD1_D3
 - Pin 10: SD1_D4
 - Pin 11: GND
 - Pin 12: SD2_D1
 - Pin 13: SD2_D2
 - Pin 14: GND
 - Pin 15: SD2_D3
 - Pin 16: SD2_D4

Additional components shown include a 33R resistor (R1415) connected to the SD_CLK_OUT signal and a TSW-110-07-G-D connector.

8 modulator data inputs (SD1/SD2 D1–D4) and one clock: XCLKOUT out, returned clock to all eight SDx_Cy inputs. 3.3 V CMOS; for an adjacent sensing board only.

8 modulator data inputs (SD1/SD2 D1–D4) and one clock: XCLKOUT out, returned clock to all eight SDx_Cy inputs. 3.3 V CMOS; for an adjacent sensing board only.